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**MediSync AI: Intelligent Hospital Resource and Patient Flow Management
SystemAgile Sprint Planning & User Story Workshop Report**

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AGILE SPRINT PLANNING & USER STORY WORKSHOP REPORT

ABSTRACT

This document presents the Agile Sprint Planning and User Story Workshop prepared for the Software Engineering course assessment. The workshop applies Agile methodologies to manage software projects effectively by focusing on iterative development, stakeholder collaboration, and measurable outcomes. It begins with defining the project vision and stakeholders, ensuring that the objectives are clear and aligned with user expectations. Next, the requirements are broken down into epics and user stories, written in the Agile format to capture functional needs from the user's perspective. Each user story is supported by acceptance criteria, framed in the Given–When–Then style to make them specific, testable, and measurable.

The document also emphasizes product backlog prioritization, where stories are organized based on business value, customer needs, and dependencies. From this backlog, a sprint backlog is created, selecting high-priority items and breaking them into actionable tasks for the development team. To ensure realistic planning, effort estimation is carried out using story points, justified through techniques like the Fibonacci sequence or Planning Poker. Finally, a sprint goal and deliverables are defined, providing clarity on what the team aims to achieve within the sprint. By covering all these aspects, the workshop demonstrates how Agile practices foster adaptability, transparency, and continuous improvement, ultimately guiding the team toward delivering valuable software features within the given timeframe.

1.INTRODUCTION

One of the major problems in software engineering projects is the lack of clear planning, structured requirements, and effective stakeholder communication. Many projects fail because objectives are not properly defined, user needs are misunderstood, and tasks are not prioritized, leading to delays and poor quality outcomes. Traditional project management methods often struggle to adapt to changing customer demands, which creates inefficiency and reduces overall satisfaction.

The purpose of this Agile Sprint Planning and User Story Workshop is to solve these challenges by applying Agile methodologies. Agile provides a flexible, iterative approach that

ensures continuous collaboration, prioritization of customer value, and measurable progress. Through this workshop, the project team defines the **project vision and stakeholders**, creates **epics and user stories**, sets **acceptance criteria**, prioritizes the **product backlog**, prepares the **sprint backlog**, estimates effort using **story points**, and establishes a **sprint goal with deliverables**.

2. PROJECT VISION & STAKEHOLDERS

Project Vision

The vision of this project is to design and deliver a secure, user-friendly, and scalable software solution that directly addresses customer needs while ensuring long-term sustainability. The project aims to provide a seamless user experience, strong data protection, and features that add measurable value to end users. By adopting Agile methodologies, the project will emphasize iterative development, continuous stakeholder collaboration, and adaptability to evolving requirements. This vision ensures that the software not only meets current expectations but also remains flexible enough to accommodate future growth and technological advancements.

The project is guided by the principle of customer-centric development, where every feature is designed to solve real problems faced by users. The vision also emphasizes transparency in communication, efficiency in execution, and quality in delivery. By aligning business objectives with user expectations, the project seeks to create a product that is both technically robust and practically useful.

Objectives

To achieve this vision, the project sets out the following objectives:

- Deliver a simple and intuitive user interface that enhances usability.
- Ensure robust security and privacy measures to protect sensitive data.
- Build a scalable architecture capable of supporting future expansion.
- Prioritize features that maximize business value and customer satisfaction.
- Maintain transparency, collaboration, and adaptability throughout the development cycle.

These objectives serve as measurable targets that guide the team's efforts during sprint planning and execution. They ensure that every sprint contributes meaningfully to the overall vision and keeps the project aligned with stakeholder expectations.

Stakeholders

Stakeholders play a critical role in shaping the success of the project. Each stakeholder has unique responsibilities and influence, and their collaboration ensures that the project remains balanced and effective.

- **Product Owner:** Acts as the voice of the customer and business. Responsible for defining requirements, prioritizing the product backlog, and ensuring that features deliver maximum value.

- **Scrum Master:** Facilitates Agile practices, removes obstacles, and ensures smooth sprint execution. Plays a key role in maintaining team productivity and adherence to Agile principles.
- **Development Team:** Composed of designers, developers, and engineers who build the product. They transform user stories into working software and ensure technical quality.
- **QA/Test Engineers:** Validate features against acceptance criteria, conduct testing, and guarantee reliability. Their role ensures that deliverables meet quality standards before release.
- **End Users/Customers:** Provide feedback, test usability, and validate whether the product meets their needs. Their satisfaction ultimately determines the success of the project.
- **Project Manager:** Oversees overall progress, manages resources, and ensures timely delivery. Provides strategic oversight and ensures that the project stays on track.

3. EPICS AND USER STORIES

Epic	User Stories
Epic 1	User Authentication
Epic 2	Dashboard Management
Epic 3	Notifications and Alerts
Epic 4	Reporting and Analytics
Epic 5	Settings and Preferences

Epics

Epics represent large features or modules of the project that can be broken down into smaller, manageable user stories. They provide a high-level view of the system's functionality and help in organizing requirements. For this project, the following epics have been identified:

1. **User Authentication** – Secure login and registration system to protect user data.
2. **Dashboard Management** – A personalized dashboard that allows users to access relevant features quickly.
3. **Notifications and Alerts** – A system to notify users about important updates and activities.
4. **Reporting and Analytics** – Tools to generate reports and provide insights into user activity.
5. **Settings and Preferences** – Options for users to customize their experience.

User Stories

Each epic is broken down into user stories that describe specific requirements from the user's perspective. These stories follow the Agile format: *As a <user>, I want <feature>, so that <benefit>.*

- **Epic 1: User Authentication**
 - *As a user, I want to register with my email and password, so that I can create a secure account.*

- *As a user, I want to log in securely, so that my personal data is protected.*
- *As a user, I want to reset my password, so that I can regain access if I forget it.*
- **Epic 2: Dashboard Management**
 - *As a user, I want a personalized dashboard, so that I can quickly access the features I use most.*
 - *As a user, I want to view my recent activity on the dashboard, so that I can track my progress.*
- **Epic 3: Notifications and Alerts**
 - *As a user, I want to receive real-time notifications, so that I stay updated on important events.*
 - *As a user, I want to manage notification preferences, so that I only receive alerts relevant to me.*
- **Epic 4: Reporting and Analytics**
 - *As a user, I want to generate activity reports, so that I can analyze my performance.*
 - *As a manager, I want to view system usage analytics, so that I can make informed decisions.*
- **Epic 5: Settings and Preferences**
 - *As a user, I want to change my profile settings, so that I can update my personal information.*
 - *As a user, I want to customize my theme preferences, so that I can improve my user experience.*

4. ACCEPTANCE CRITERIA

Epic 1: User Authentication

- **User Story:** *As a user, I want to log in securely, so that my personal data is protected.*

- *Given* a registered user with valid credentials,
- *When* they enter their username and password,
- *Then* they should be logged in successfully and redirected to the dashboard.
- **User Story:** *As a user, I want to reset my password, so that I can regain access if I forget it.*
 - *Given* a user who has forgotten their password,
 - *When* they request a password reset,
 - *Then* they should receive a secure reset link via email and be able to set a new password.

Epic 2: Dashboard Management

- **User Story:** *As a user, I want a personalized dashboard, so that I can quickly access the features I use most.*
 - *Given* a logged-in user,
 - *When* they open the dashboard,
 - *Then* they should see personalized content based on their profile and recent activity.
- **User Story:** *As a user, I want to view my recent activity on the dashboard, so that I can track my progress.*
 - *Given* a logged-in user,
 - *When* they access the dashboard,
 - *Then* they should see a list of their most recent actions displayed clearly.

Epic 3: Notifications and Alerts

- **User Story:** *As a user, I want to receive real-time notifications, so that I stay updated on important events.*
 - *Given* a logged-in user,
 - *When* a new event occurs,
 - *Then* a notification should appear instantly on their dashboard or device.

- **User Story:** *As a user, I want to manage notification preferences, so that I only receive alerts relevant to me.*
 - Given a logged-in user,
 - When they update their notification settings,
 - Then only the selected types of notifications should be delivered.

Epic 4: Reporting and Analytics

- **User Story:** *As a user, I want to generate activity reports, so that I can analyze my performance.*
 - Given a logged-in user,
 - When they request a report,
 - Then the system should generate and display a detailed activity report.
- **User Story:** *As a manager, I want to view system usage analytics, so that I can make informed decisions.*
 - Given a manager with access rights,
 - When they open the analytics dashboard,
 - Then they should see accurate charts and statistics about system usage.

Epic 5: Settings and Preferences

- **User Story:** *As a user, I want to change my profile settings, so that I can update my personal information.*
 - Given a logged-in user,
 - When they edit their profile details,
 - Then the updated information should be saved and displayed correctly.
- **User Story:** *As a user, I want to customize my theme preferences, so that I can improve my user experience.*
 - Given a logged-in user,
 - When they select a new theme,

5. PRODUCT BACKLOG PRIORITIZATION

Priority Level	Epic / Feature	User Story Example	Reason for Priority
High	User Authentication	<i>As a user, I want to log in securely, so that my personal data is protected.</i>	Essential for system security and access control.
High	Dashboard Management	<i>As a user, I want a personalized dashboard, so that I can quickly access features.</i>	Core functionality for user experience.
Medium	Notifications & Alerts	<i>As a user, I want to receive of real-time notifications, so that I stay updated.</i>	Improves usability but not critical for initial release.
Medium	Reporting & Analytics	<i>As a manager, I want to view system usage analytics, so that I can make decisions.</i>	Adds business value but depends on core features.
Low	Settings & Preferences	<i>As a user, I want to customize my theme preferences, so that I can improve UX.</i>	Enhances personalization, can be delivered later.

Explanation

- **High Priority:** Authentication and dashboard are critical for the system to function and deliver immediate value.
- **Medium Priority:** Notifications and reporting add usability and insights but can be implemented after core features.
- **Low Priority:** Settings and preferences improve personalization but are not essential for the first sprint.

6.SPRINT BACKLOG TABLE

User Story	Development Tasks
Login Securely	- Design login page UI (username, password fields, buttons) - Implement backend logic for authentication - Set up secure password storage (hashing/encryption) - Create error handling for invalid login attempts - Develop password reset functionality with email verification - Write unit tests for authentication module
Personalized Dashboard	- Design dashboard layout (navigation bar, widgets, sections) - Implement personalized content display based on user profile - Connect dashboard to backend database for retrieving user data - Create recent activity tracking feature - Add role-based access control for different users - Write integration tests for dashboard functionality

Sprint Goal

Deliver a working prototype with **secure login functionality** and a **basic personalized dashboard**.

Expected Deliverables

- Secure login module with password reset option.
- Functional dashboard interface with personalized content.
- Database integration for authentication and dashboard data.
- Unit and integration test results ensuring reliability.

7. STORY POINT ESTIMATION

User Story	Story Points	Justification
Login Securely (<i>As a user, I want to log in securely, so that my personal data is protected.</i>)	8	Requires backend authentication logic, secure password storage, error handling, and testing. High complexity and critical for system security.
Password Reset (<i>As a user, I want to reset my password, so that I can regain access if I forget it.</i>)	5	Involves email verification and secure reset link generation. Moderate complexity with dependency on authentication.
Personalized Dashboard (<i>As a user, I want a personalized dashboard, so that I can quickly access the features I use most.</i>)	8	Needs UI design, backend integration, and personalized content logic. High effort due to multiple components.
Recent Activity Tracking (<i>As a user, I want to view my recent activity on the dashboard, so that I can track my progress.</i>)	3	Relatively simple feature that retrieves and displays user activity from the database. Lower complexity.
Notifications (<i>As a user, I want to receive real-time notifications, so that I stay updated on important events.</i>)	5	Requires event triggers, backend logic, and real-time updates. Moderate complexity.
Settings & Preferences (<i>As a user, I want to customize my theme preferences, so that I can improve my user experience.</i>)	2	Simple UI customization with minimal backend logic. Low complexity.

Story points are assigned using the **Fibonacci sequence (1, 2, 3, 5, 8, 13...)** to reflect complexity, effort, and risk. Lower points indicate simpler tasks, while higher points represent more complex or time-consuming work.

8. SPRINT GOAL & EXPECTED DELIVERABLES

Sprint Goal

The goal of Sprint 1 is to deliver a **working prototype** that establishes the foundation of the system. This includes implementing **secure user authentication** and a **basic personalized dashboard**. By achieving this, the team ensures that users can log in safely and access a functional interface tailored to their needs. The sprint focuses on building essential features that provide immediate business value and customer satisfaction.

Deliverable	Description
Secure Login Module	A fully functional login system with credential validation, password encryption, and error handling.
Password Reset Feature	A secure password recovery process using email verification and reset links.
Basic Dashboard Interface	A personalized dashboard layout with navigation, widgets, and user-specific content.
Recent Activity Tracking	A feature that displays the user's latest actions for better visibility and progress tracking.
Database Integration	Backend connections to store and retrieve user authentication and dashboard data.
Testing Results	Unit tests for authentication and integration tests for dashboard functionality to ensure reliability.

9. Discussion

Agile methodology helps solve common project challenges by focusing on clarity, collaboration, and adaptability. In this project, requirements were broken into epics and user stories, making them manageable. The use of acceptance criteria ensured testability, while backlog prioritization focused on delivering high-value features first. The sprint backlog and story points estimation organized tasks effectively, ensuring balanced workload and realistic planning.

10 Conclusion

Agile Sprint Planning ensures that projects deliver customer-focused outcomes by breaking down complex requirements into achievable tasks. Sprint 1 will deliver a secure login system and a personalized dashboard, forming the foundation for future enhancements. Overall, Agile provides structure, flexibility, and quality assurance, empowering teams to adapt quickly and deliver software that meets evolving user needs.



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